1 Introduction to the density and the reduced density matrix

1.1 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i, i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \,, \qquad p_i \in \mathbb{R} \;\; \& \;\; \sum_i p_i = 1, \tag{1}$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

1.2 Observables on a composite system

Take a system A with some observable M, and a system B, giving the composite system AB (see Nielsen & Chuang Box 2.6). Assume we can measure M on system A; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle\,|\psi\rangle$ with $M\,|m\rangle = m\,|m\rangle$. Applying the measurement to $|m\rangle\,|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m\, P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB. □

1.3 The reduced density matrix as a partial trace

Measurement averages[1] must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A, i.e.

$$\langle \hat{M} \rangle = \mathrm{tr}\big(M\rho^A\big) \overset{!}{=} \mathrm{tr}\big(\tilde{M}\rho^{AB}\big) = \mathrm{tr}\big((M \otimes \mathbb{1}_B)\rho^{AB}\big). \tag{2}$$

Take $\rho^A = \mathrm{tr}_B(\rho^{AB})$ where

$$\mathrm{tr}_B\big(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|\big) \equiv |a_1\rangle\langle a_2| \; \mathrm{tr}\big(|b_1\rangle\langle b_2|\big) \qquad \& \qquad \mathrm{tr}_B(\cdot) \text{ linear}. \tag{3}$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M\,|i\rangle = m_i\,|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha \, |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijk\ell} c^\alpha_{ij} (c^\alpha_{k\ell})^* \; |i\rangle_A \, |j\rangle_B \, \langle\ell|_B \, \langle k|_A \,, \tag{4}$$

$$\rho^A = \sum_\alpha \sum_{ijk\ell} p^\alpha c^\alpha_{ij} (c^\alpha_{k\ell})^* \; |i\rangle\langle k|_A \underbrace{\mathrm{tr}\big(|j\rangle\langle\ell|_B\big)}_{=\delta_{j\ell}} = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij} (c^\alpha_{kj})^* \; |i\rangle\langle k|_A \,. \tag{5}$$

Therefore

$$\mathrm{tr}\big(M\rho^A\big) = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij} (c^\alpha_{kj})^* \underbrace{\mathrm{tr}\big(m_i \, |i\rangle\langle k|_A\big)}_{=m_i\delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i \, c^\alpha_{ij} (c^\alpha_{ij})^*, \tag{6}$$

[1] The expectation value for a mixed state can be written as $\langle \hat{O} \rangle \equiv \mathrm{tr}\big(\rho \hat{O}\big) = \sum_i p_i \, \langle\psi_i|\, \hat{O} \, |\psi_i\rangle$.

and similarly $\mathrm{tr}\big((M \otimes \mathbb{1}_B)\rho^{AB}\big) = \sum_{\alpha,ij} p^\alpha m_i \, c_{ij}^\alpha (c_{ij}^\alpha)^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Nielsen & Chuang Box 2.6).

1.4 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^* \; |i\rangle_A \, |j\rangle_B \, \langle\ell|_B \, \langle k|_A \,, \tag{7}$$

$$\rho_A = \mathrm{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^* \; |i\rangle\langle k|_A \,, \tag{8}$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \tag{9}$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A \times d_B}$, i.e. $|\psi\rangle = \sum_{ij} c_{ij} \, |i\rangle_A \, |j\rangle_B$ with $\sum_{ij} |c_{ij}|^2 = 1$.

Furthermore, eq. (8) results in

$$\begin{aligned} \mathrm{tr}\, \rho_A &= \mathrm{tr} \left[\sum_{ijk} c_{ij}(c_{kj})^* \, |i\rangle\langle k|_A \right] \\ &= \sum_{ijkl} c_{ij}(c_{kj})^* \delta_{il}\delta_{kl} \\ &= \sum_{jl} c_{lj}(c_{lj})^* = 1, \end{aligned}$$

which means that that the eigenvalues of the reduced density matrix sum to one. This can be seen using that any hermitian matrix can be diagonalized and the cylic property of the trace

$$\mathrm{tr}\, \rho_A = \mathrm{tr}[ODO^\dagger] = \mathrm{tr}\, D = \sum_i \lambda_i = 1. \tag{10}$$

1.5 Connection to the SVD

Now use the SVD on ψ:

$$\psi = USV^\dagger \qquad \text{with } U \in \mathbb{C}^{d_A \times d_A},\ V \in \mathbb{C}^{d_B \times d_B} \text{ unitary.} \tag{11}$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger V S^\dagger U^\dagger = USS^\dagger U^\dagger, \qquad SS^\dagger = \mathrm{diag}(s_1^2, \ldots, s_r^2, 0, \ldots, 0), \tag{12}$$

and write $S^2 = \mathrm{diag}(s_1^2, \ldots, s_r^2)$. Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = ODO^\dagger = US^2U^\dagger$ and $D = \mathrm{diag}(\lambda_1, \ldots)$, so that

$$\boxed{\lambda_i = s_i^2}, \tag{13}$$

i.e. the eigenvalues are given by the squared Schmidt values and hence non-negative. Thus, we can conclude together with eq. (10) that

$$\boxed{0 \leq \lambda_i \leq 1}. \tag{14}$$

1.6 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).